# SRIRAM POSIMSETTI

AI/ML Engineer | LLM Pipelines | RAG Systems | Generative AI

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## PROFESSIONAL SUMMARY

AI/ML Engineer with 1+ year of hands-on experience building, evaluating, and optimizing LLM-based pipelines, RAG systems, and document processing workflows for production-grade applications. Proficient in Python, Hugging Face Transformers, LangChain, and OpenAI LLMs with strong expertise in embeddings, vector databases (FAISS), and prompt engineering. Experienced in running experiments to optimize accuracy, latency, and cost, and in developing evaluation datasets and validation frameworks for AI systems.

## TECHNICAL SKILLS

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| **Languages** | Python, JavaScript, SQL |
| **LLM & Gen AI** | OpenAI LLMs, Hugging Face Transformers, LangChain, LangGraph, RAG Architecture, Prompt Engineering, RLHF, Agentic AI, Instruction Tuning, Embeddings |
| **ML Frameworks** | TensorFlow, Keras, PyTorch, Scikit-learn |
| **Vector Databases** | FAISS, Semantic Search, Similarity Retrieval |
| **Doc Processing** | Multi-document ingestion pipelines, PDF parsing, text chunking, embedding generation |
| **Evaluation** | ROUGE scoring, human evaluation benchmarks, accuracy/latency/cost optimization, validation framework design |
| **NLP** | Text classification, text mining, summarization, NER, BioBERT, pattern recognition |
| **APIs & Backend** | FastAPI, Django REST Framework, Flask, Streamlit |
| **Databases** | PostgreSQL, MySQL |
| **Computer Vision** | CNN, OpenCV, Transfer Learning, TensorFlow/Keras, Image Classification, Object Detection |
| **Tools** | Git, GitHub, Postman, VSCode, Power BI |

## PROFESSIONAL EXPERIENCE

### AI/ML Engineer & Full Stack Developer

*June 2024 – Present*

**Ecode Learning Software Services | Hyderabad, India**

- Evaluate and improve LLM-based pipelines by benchmarking model outputs across accuracy, latency, and cost dimensions; iterate on prompt strategies to optimize performance.
- Build and optimize RAG (Retrieval-Augmented Generation) systems and document processing pipelines using LangChain, FAISS, and OpenAI LLMs for enterprise knowledge management.
- Run systematic experiments comparing embedding models, chunking strategies, and retrieval configurations to reduce hallucination rates and improve contextual relevance.
- Develop evaluation datasets and validation frameworks to measure LLM output quality; implement automated testing pipelines for production AI systems.
- Fine-tune transformer-based LLMs using Hugging Face on domain-specific datasets, applying instruction tuning and RLHF-ready alignment strategies.
- Integrate Generative AI and Agentic AI workflows (LangGraph) into scalable production applications deployed on AWS (EC2, S3, Lambda) with CI/CD pipelines.
- Design and expose ML model inference endpoints via FastAPI for real-time, low-latency LLM-powered application features.

### Machine Learning Intern

*June 2023 – July 2023*

**SAK Informatics | India**

- Assisted in building and evaluating supervised ML models for classification and regression tasks on real-world large datasets.
- Performed data preprocessing, feature engineering, and exploratory data analysis to prepare model training pipelines.
- Ran model evaluation experiments tracking accuracy, precision, recall, and F1-score metrics to assess and improve model quality.

## EDUCATION

### Bachelor of Technology (B.Tech) — Electrical & Electronics Engineering

*Graduated: 2024*

Y.S.R. Engineering College of Yogi Vemana University, Kadapa

## KEY PROJECTS

### Multi-Document Intelligence System Using RAG (Retrieval-Augmented Generation)

*Domain: Generative AI | LLM Pipelines | Enterprise Document Processing*

*Tech Stack: Python, LangChain, FAISS, OpenAI LLMs, FastAPI, Streamlit, PostgreSQL, Docker*

- Designed and built end-to-end RAG pipelines covering document ingestion, text chunking, embedding generation (OpenAI), vector storage in FAISS, and semantic retrieval for LLM-grounded question answering.
- Ran experiments to optimize chunking strategy, embedding model selection, and retrieval top-k parameters, improving answer relevance and reducing pipeline latency.
- Developed evaluation datasets and validation frameworks to benchmark RAG output accuracy, factuality, and retrieval precision across diverse document types.
- Engineered prompt templates to minimize hallucinations and maximize factual grounding; deployed via FastAPI and Streamlit on AWS with Docker containerization.

### Clinical Decision Support System Using Vector DB & LLMs

*Domain: Healthcare Generative AI | NLP | LLM Pipelines*

*Tech Stack: Python, FAISS, LangChain, BioBERT, LLMs, FastAPI*

- Built a domain-specific LLM pipeline for healthcare including medical document embedding (BioBERT), FAISS-based semantic retrieval, and LLM-generated evidence-backed clinical responses.
- Evaluated pipeline outputs using domain-specific validation frameworks, ensuring traceable and explainable AI responses linked to source medical literature.
- Developed FastAPI inference endpoints for real-time clinical query handling with low-latency, high-reliability LLM responses in production environments.

### Multi-Disease Prediction System Using Medical Imaging

*Domain: Deep Learning | Computer Vision | Healthcare AI*

*Tech Stack: Python, CNN, TensorFlow/Keras, OpenCV, Transfer Learning*

- Designed and trained CNN architectures with transfer learning (ResNet, VGG) for multi-class medical image classification covering pneumonia, brain tumors, and diabetic retinopathy.
- Ran model evaluation experiments using accuracy, precision, recall, F1-score, and AUC-ROC metrics; iterated on architecture and data augmentation strategies to improve performance.
- Deployed trained models via inference APIs for real-time diagnostic assistance, integrating OpenCV-based image preprocessing pipelines.

## CORE COMPETENCIES

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|---|---|---|---|
| LLM Pipeline Development | RAG System Design | Document Processing | Embeddings & Vector Search |
| Prompt Engineering | Model Fine-Tuning | RLHF | Hugging Face Transformers |
| LangChain & LangGraph | OpenAI LLMs | Evaluation Dataset Development | Validation Frameworks |
| Accuracy & Latency Optimization | Generative AI | Agentic AI | NLP |
| Python | AWS | FastAPI | Production AI Systems |